

```
gayathri@Gayathri: ~/mini_parser
gayathri@Gayathri:~$ mkdir mini_parser
gayathri@Gayathri:~$ cd mini_parser
gayathri@Gayathri:~/mini_parser$ pwd
/home/gayathri/mini_parser
gayathri@Gayathri:~/mini_parser$ nano parser.c
gayathri@Gayathri:~/mini_parser$ ls
parser.c
gayathri@Gayathri:~/mini_parser$ gcc -g parser.c -o parser
gayathri@Gayathri:~/mini_parser$ ls
parser  parser.c
gayathri@Gayathri:~/mini_parser$ ./parser
=====
      MINI SHELL TOKENIZER
      AND PARSER
=====

Enter a command:
> ls -l

===== TOKENIZER OUTPUT =====
Token 1: WORD      Value: ls
Token 2: WORD      Value: -l

===== VALIDATION =====
Syntax validation PASSED.

===== PARSER / EXECUTION STRUCTURE =====

Command 1:
Arguments: [ls] [-l]
Pipe: NO

Memory cleanup completed.
gayathri@Gayathri:~/mini_parser$ ./parser
=====
```

```
gayathri@Gayathri: ~/mini_parser
Enter a command:
> ls      -l      -a

===== TOKENIZER OUTPUT =====
Token 1: WORD      Value: ls
Token 2: WORD      Value: -l
Token 3: WORD      Value: -a

===== VALIDATION =====
Syntax validation PASSED.

===== PARSER / EXECUTION STRUCTURE =====

Command 1:
Arguments: [ls] [-l] [-a]
Pipe: NO

Memory cleanup completed.
gayathri@Gayathri:~/mini_parser$ ./parser
=====
      MINI SHELL TOKENIZER
      AND PARSER
=====

Enter a command:
> ls -l | grep .c

===== TOKENIZER OUTPUT =====
Token 1: WORD      Value: ls
Token 2: WORD      Value: -l
Token 3: PIPE      Value: |
Token 4: WORD      Value: grep
Token 5: WORD      Value: .c
```

```

gayathri@Gayathri: ~/mini_parser
===== VALIDATION =====
Syntax validation PASSED.

===== PARSER / EXECUTION STRUCTURE =====

Command 1:
Arguments: [ls] [-l]
Pipe: YES

Command 2:
Arguments: [grep] [.]
Pipe: NO

Memory cleanup completed.
gayathri@Gayathri:~/mini_parser$ ls -l | grep .c
-rw-r--r-- 1 gayathri gayathri 11403 Aug 29 10:00 parser.c
gayathri@Gayathri:~/mini_parser$ ./parser
=====
      MINI SHELL TOKENIZER
      AND PARSER
=====

Enter a command:
> ls > output.txt

===== TOKENIZER OUTPUT =====
Token 1: WORD           Value: ls
Token 2: REDIRECT_OUTPUT Value: >
Token 3: WORD           Value: output.txt

===== VALIDATION =====
Syntax validation PASSED.

===== PARSER / EXECUTION STRUCTURE =====

```

```

gayathri@Gayathri: ~/mini_parser

Command 1:
Arguments: [ls]
Output file: output.txt
Output mode: OVERWRITE
Pipe: NO

Memory cleanup completed.
gayathri@Gayathri:~/mini_parser$ ./parser
=====
      MINI SHELL TOKENIZER
      AND PARSER
=====

Enter a command:
> sort < input.txt

===== TOKENIZER OUTPUT =====
Token 1: WORD           Value: sort
Token 2: REDIRECT_INPUT Value: <
Token 3: WORD           Value: input.txt

===== VALIDATION =====
Syntax validation PASSED.

===== PARSER / EXECUTION STRUCTURE =====

Command 1:
Arguments: [sort]
Input file: input.txt
Pipe: NO

Memory cleanup completed.
gayathri@Gayathri:~/mini_parser$ ./parser
=====

```

```
gayathri@Gayathri: ~/mini_pi x + v
=====
MINI SHELL TOKENIZER
AND PARSER
=====

Enter a command:
> echo hello >> output.txt

===== TOKENIZER OUTPUT =====
Token 1: WORD          Value: echo
Token 2: WORD          Value: hello
Token 3: REDIRECT_APPEND Value: >>
Token 4: WORD          Value: output.txt

===== VALIDATION =====
Syntax validation PASSED.

===== PARSER / EXECUTION STRUCTURE =====

Command 1:
Arguments: [echo] [hello]
Output file: output.txt
Output mode: APPEND
Pipe: NO

Memory cleanup completed.
gayathri@Gayathri:~/mini_parser$ ./parser
=====
MINI SHELL TOKENIZER
AND PARSER
=====

Enter a command:
>
```

```
gayathri@Gayathri: ~/mini_pi x + v

===== TOKENIZER OUTPUT =====
No tokens found.

===== VALIDATION =====
ERROR: Empty command.
Syntax validation FAILED.
gayathri@Gayathri:~/mini_parser$ ./parser
=====
MINI SHELL TOKENIZER
AND PARSER
=====

Enter a command:
> | ls

===== TOKENIZER OUTPUT =====
Token 1: PIPE          Value: |
Token 2: WORD          Value: ls

===== VALIDATION =====
ERROR: Command cannot start with '|'.
Syntax validation FAILED.
gayathri@Gayathri:~/mini_parser$ ./parser
=====
MINI SHELL TOKENIZER
AND PARSER
=====

Enter a command:
> ls |

===== TOKENIZER OUTPUT =====
Token 1: WORD          Value: ls
Token 2: PIPE          Value: |
```

```
gayathri@Gayathri: ~/mini_pi x + v
===== VALIDATION =====
ERROR: Command cannot end with '|'.
Syntax validation FAILED.
gayathri@Gayathri:~/mini_parser$ ./parser
=====
      MINI SHELL TOKENIZER
      AND PARSER
=====

Enter a command:
> ls || grep

===== TOKENIZER OUTPUT =====
Token 1: WORD           Value: ls
Token 2: PIPE           Value: |
Token 3: PIPE           Value: |
Token 4: WORD           Value: grep

===== VALIDATION =====
ERROR: Empty command between pipes.
Syntax validation FAILED.
gayathri@Gayathri:~/mini_parser$ ./parser
=====
      MINI SHELL TOKENIZER
      AND PARSER
=====

Enter a command:
> cat input.txt | grep hello > result.txt

===== TOKENIZER OUTPUT =====
Token 1: WORD           Value: cat
Token 2: WORD           Value: input.txt
Token 3: PIPE           Value: |
Token 4: WORD           Value: grep
```

```
gayathri@Gayathri: ~/mini_pi x + v
gayathri@Gayathri:~/mini_parser$ ./parser
=====
      MINI SHELL TOKENIZER
      AND PARSER
=====

Enter a command:
> cat input.txt | grep hello > result.txt

===== TOKENIZER OUTPUT =====
Token 1: WORD           Value: cat
Token 2: WORD           Value: input.txt
Token 3: PIPE           Value: |
Token 4: WORD           Value: grep
Token 5: WORD           Value: hello
Token 6: REDIRECT_OUTPUT Value: >
Token 7: WORD           Value: result.txt

===== VALIDATION =====
Syntax validation PASSED.

===== PARSER / EXECUTION STRUCTURE =====

Command 1:
Arguments: [cat] [input.txt]
Pipe: YES

Command 2:
Arguments: [grep] [hello]
Output file: result.txt
Output mode: OVERWRITE
Pipe: NO

Memory cleanup completed.
gayathri@Gayathri:~/mini_parser$ |
```